Week 5: aquatic invertebrates

Individual assignment 5

Erick Morales Oyola

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1. Set up

Packages

```
#Loading general use and cleaning packages
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Loading package for wrapping long axis labels
library(scales)

#Loading package for reading Excel files
library(readxl)

#Loading package for visualizing missing data
library(naniar)

#Loading package for arranging plots
library(patchwork)
```

Data

```
#Reading in taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#Reading in aquatic invertebrate data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

#Reading in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

#Reading in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
 ↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning

```
#Creating a new object from site information
ncos_sites <- sites |>
  #Filtering to only include NCOS sites sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
#Creating a clean taxon list for joining with aquatic invertebrate data
taxon_list_clean <- taxon_list |>
  #Converting taxon names to snake case so they match the aquatic
   ↪  invertebrate columns
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```
#Creating a clean water quality object
water_quality_clean <- water_quality |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only water quality surveys from quarterly NCOS sites
  semi_join(ncos_sites, by = "site") |>
  #Creating a date-site column for joining with aquatic invertebrate data
  unite("date_site", date, site, remove = TRUE) |>
  #Grouping by date-site survey
```

```
  group_by(date_site) |>
  #Calculating median pH, salinity, and dissolved oxygen for each survey
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  #Ungrouping the dataframe
  ungroup() |>
  #Filtering out the salinity outlier identified in class
  filter(date_site != "2022-08-12_NVBR")
```

```
#Creating a clean aquatic invertebrate object
aq_ins_clean <- aq_ins |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only sites that occur in the NCOS sites object
  semi_join(ncos_sites, by = c("site")) |>
  #Renaming the vial date column to date
  rename(date = date_on_vial) |>
  #Selecting survey information and focal aquatic invertebrate taxa
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  #Keeping only filtered beaker samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  #Converting taxon columns into long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  #Grouping by date, site, and taxon
  group_by(date, site, taxon) |>
  #Calculating average abundance per liter from the 7.5 liter sample volume
  summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
  #Ungrouping the dataframe
  ungroup() |>
  #Creating a date-site column for joining with water quality data
```

```
unite("date_site", date, site, remove = FALSE) |>
#Joining water quality information to each survey
left_join(water_quality_clean, by = c("date_site")) |>
#Joining taxonomic information to each taxon
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
#Adding full site names for clearer figure labels
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~  "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
#Ordering sites by median salinity so site patterns are easier to compare
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
 ↪  = TRUE),
       site_full = fct_rev(site_full))
```

Problems

1. Visualizing differences across sites in major

taxon abundance

a. Plan your figure

- X-Axis: `site_full`
- Y-Axis: log + 1 transformed average Ostracod abundance per liter
- Geometry to represent average abundance/L: `geom_jitter()`
- Geometry to represent medians: `stat_summary()`

b. Wrangle your data

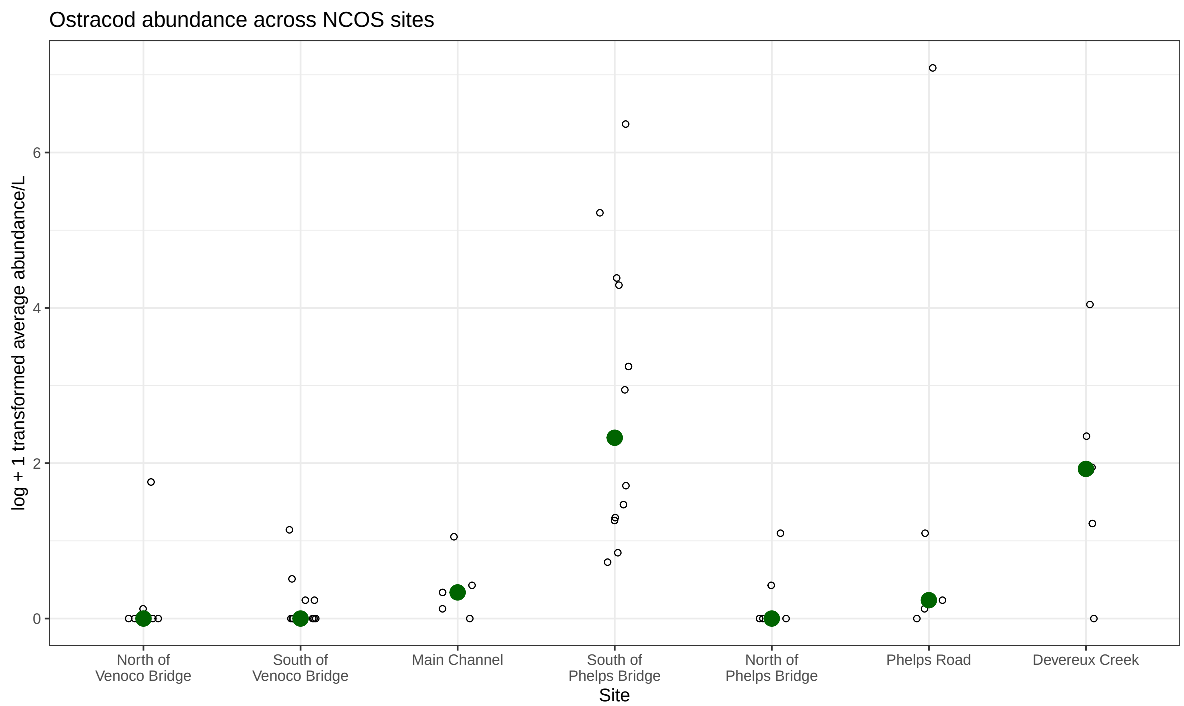
```
#Creating a new object from clean aquatic invertebrate data
ostracods <- aq_ins_clean |>
  #Filtering to only include ostracods
  filter(taxon == "ostracod") |>
  #Log + 1 transforming average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

#Displaying 10 random rows from the ostracod data frame
ostracods |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-11-12_NMC  2023-11-12 00:00:00 NMC   ostr~   0.533  NA      47       14.0
 2 2023-09-12_NVBR 2023-09-12 00:00:00 NVBR  ostr~   0        9.14  NA       20.3
 3 2021-08-25_NPB  2021-08-25 00:00:00 NPB   ostr~ 581.       7.52   4.57    4.5
 4 2022-02-23_NPB1 2022-02-23 00:00:00 NPB1  ostr~   0      NA      NA       NA
 5 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1  ostr~   0      NA      NA       NA
 6 2023-05-04_NMC  2023-05-04 00:00:00 NMC   ostr~   1.87     8.32  NA       10.3
 7 2022-05-05_NEC  2022-05-05 00:00:00 NEC   ostr~   0.267    9.45  63.8      7.3
 8 2023-02-21_NEC  2023-02-21 00:00:00 NEC   ostr~   0        8.56  NA        8.8
 9 2023-11-11_NEC  2023-11-11 00:00:00 NEC   ostr~   0      NA      38.6      0.25
10 2023-03-03_NDC  2023-03-03 00:00:00 NDC   ostr~   5.73     7.45   0.688    5.64
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
#Creating an ostracod abundance figure across sites
ostracods_plot <- ggplot(data = ostracods,
                         #Setting x-axis to site and y-axis to
                          ↪ log-transformed abundance
                         mapping = aes(x = site_full,
                                       y = log_ave_lit)) +
  #Adding jittered points to show individual observations
  geom_jitter(height = 0,
              shape = 21,
```

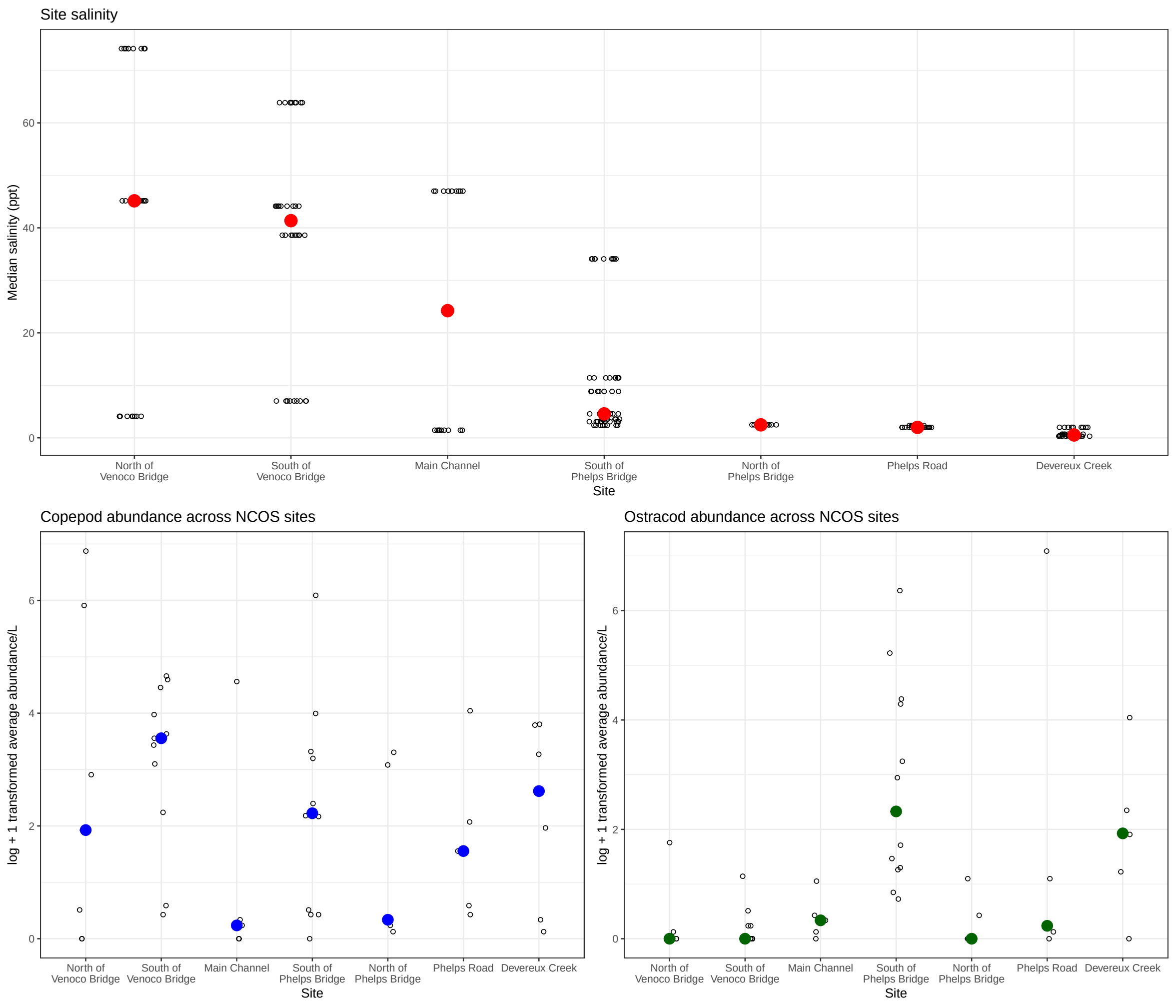
```
                width = 0.1) +
  #Adding larger points to show median ostracod abundance at each site
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "darkgreen") +
  #Wrapping long site names on the x-axis
  scale_x_discrete(labels = label_wrap(15)) +
  #Relabeling axes and adding a title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracod abundance across NCOS sites") +
  #Using a different theme from default
  theme_bw()

#Displaying ostracod abundance figure
ostracods_plot
```



d. Create a multipanel figure

```
#Creating salinity figure from clean aquatic invertebrate data
salinity <- ggplot(data = aq_ins_clean,
                   #Setting x-axis to site and y-axis to median salinity
                   mapping = aes(x = site_full,
                                 y = med_sal)) +
  #Adding jittered points to show salinity values from each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  #Wrapping long site labels on the x-axis
  scale_x_discrete(labels = label_wrap(width = 15)) +
  #Adding red points to show median salinity at each site
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  #Relabeling axes and adding a title
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  #Using a different theme from default
  theme_bw()

#Creating a clean copepod object from aquatic invertebrate data
copepods <- aq_ins_clean |>
  #Filtering to only include copepods
  filter(taxon == "copepod") |>
  #Log + 1 transforming average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
#Creating copepod abundance figure across sites
copepods_plot <- ggplot(data = copepods,
                        #Setting x-axis to site and y-axis to log-transformed
                         ↪  abundance
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
  #Adding jittered points to show individual observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  #Adding blue points to show median copepod abundance at each site
  stat_summary(geom = "point",
```

```
               fun = "median",
               size = 4,
               color = "blue") +
  #Wrapping long site labels on the x-axis
  scale_x_discrete(labels = label_wrap(15)) +
  #Relabeling axes and adding a title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepod abundance across NCOS sites") +
  #Using a different theme from default
  theme_bw()

#Combining salinity, copepod, and ostracod figures into one multipanel plot
salinity / (copepods_plot | ostracods_plot)
```



e. Interpret your figure

Copepod and Ostracod abundance differ across sites, but they do not show the exact same site-level pattern. Copepods have the highest median abundance at South of Venoco Bridge, with Devereux Creek and South of Phelps Bridge also showing relatively higher median values. Ostracods, on the other hand, appear highest at South of Phelps Bridge and Devereux Creek. These observations are visible from the larger colored median points in the lower two panels, which are highest at different sites for each taxon.

Ostracod abundance does not appear to increase simply with higher salinity. The highest Ostracod median values occur at South of Phelps Bridge and Devereux Creek, which are both lower-salinity sites compared to North and South of Venoco Bridge in the salinity panel. This suggests that Ostracod abundance may be more significantly shaped by other site-level conditions, or that its relationship with salinity is not as direct as a simple positive relationship.

2. Visualizing total abundance as a function of

salinity

a. Plan your figure

- X-Axis: `med_sal`
- Y-Axis: log + 1 transformed average abundance per liter
- Color: `taxon`
- Geometry: `geom_point()`
- Facets: `facet_wrap(~ taxon, scales = "free")`

b. Wrangle your data

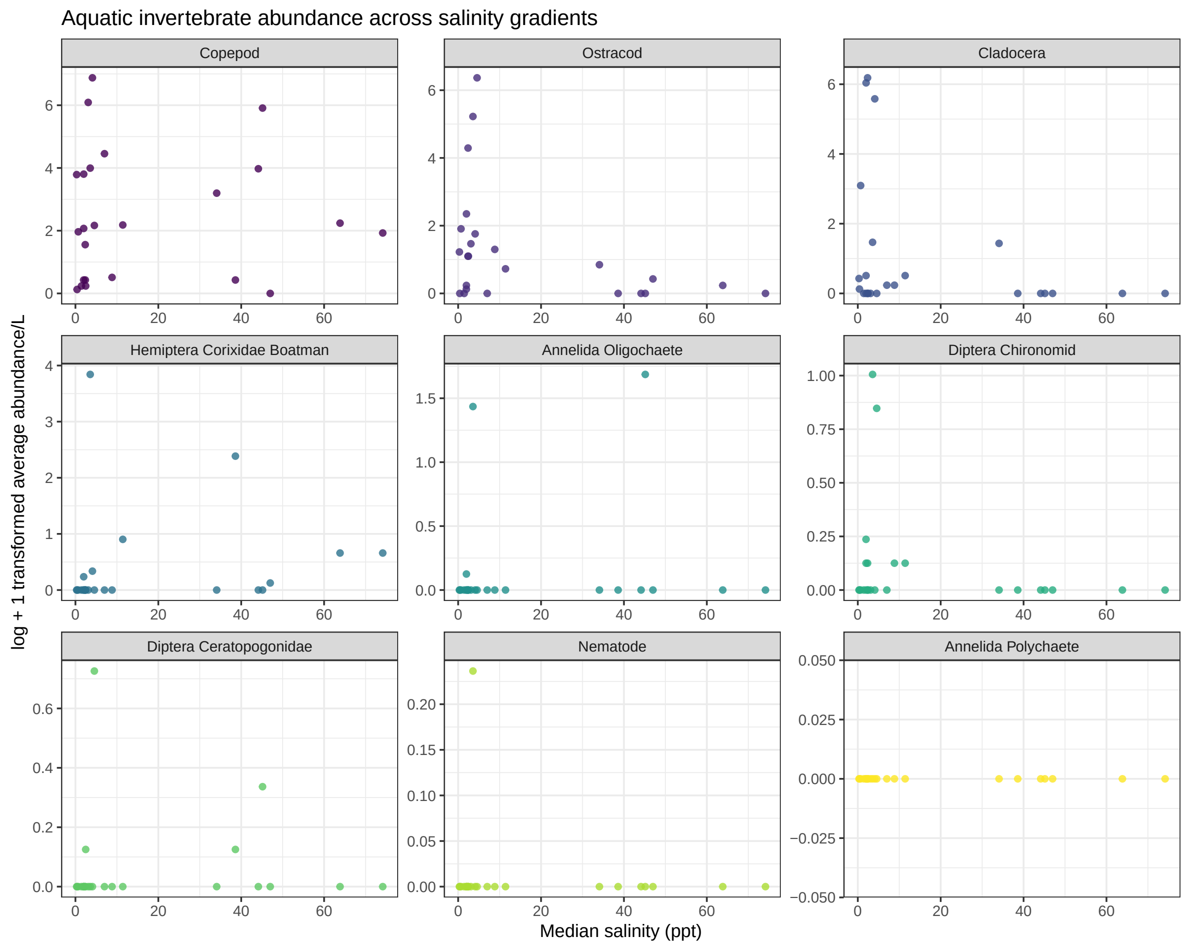
```
#Creating a data frame for visualizing abundance as a function of salinity
taxon_salinity <- aq_ins_clean |>
  #Filtering out rows without salinity values
  filter(!is.na(med_sal)) |>
  #Creating log-transformed abundance and cleaner taxon labels
  mutate(log_ave_lit = log(ave_lit + 1),
         taxon_title = str_to_title(str_replace_all(taxon, "_", " "))) |>
  #Ordering taxa by their maximum log-transformed abundance
  mutate(taxon_title = fct_reorder(taxon_title,
                                   log_ave_lit,
                                   .fun = max,
                                   .desc = TRUE)) |>
  #Keeping only the columns needed for the figure and interpretation
  select(date, site, site_full, taxon_title, ave_lit,
         log_ave_lit, med_sal)

#Displaying 10 random rows from the salinity-abundance data frame
taxon_salinity |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 7
   date                 site  site_full   taxon_title ave_lit log_ave_lit med_sal
```

```
  <dttm>              <chr> <fct>        <fct>         <dbl>        <dbl>   <dbl>
 1 2021-11-16 00:00:00 NPB   South of P~ Nematode      0            0        2.38
 2 2022-05-11 00:00:00 NDC   Devereux C~ Cladocera     0.133        0.125    0.398
 3 2024-01-28 00:00:00 NPB   South of P~ Annelida P~   0            0        3.08
 4 2022-02-25 00:00:00 NEC   South of V~ Nematode      0            0       44.1
 5 2022-05-13 00:00:00 NVBR  North of V~ Ostracod      0            0       74.2
 6 2021-11-16 00:00:00 NPB   South of P~ Diptera Ch~   0            0        2.38
 7 2022-05-12 00:00:00 NPB   South of P~ Diptera Ch~   0.133        0.125   11.4
 8 2022-11-19 00:00:00 NPB1  North of P~ Ostracod      2            1.10     2.47
 9 2023-03-03 00:00:00 NPB2  Phelps Road Diptera Ch~   0.267        0.236    1.98
10 2023-03-03 00:00:00 NDC   Devereux C~ Cladocera    21.1          3.09     0.688
```

c. Code your figure

```
#Creating a figure showing abundance as a function of salinity by taxon
ggplot(data = taxon_salinity,
       #Setting x-axis to salinity, y-axis to abundance, and color to taxon
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon_title)) +
  #Adding points to show each taxon observation
  geom_point(alpha = 0.8) +
  #Separating each taxon into its own panel with free x- and y-axis scales
  facet_wrap(~ taxon_title,
             scales = "free") +
  #Using non-default colors from the viridis palette
  scale_color_viridis_d() +
  #Relabeling axes and adding a clear title
  labs(x = "Median salinity (ppt)",
       y = "log + 1 transformed average abundance/L",
       title = "Aquatic invertebrate abundance across salinity gradients") +
  #Using a different theme from default
  theme_bw() +
  #Removing the legend because each taxon is already labeled by its panel
  theme(legend.position = "none")
```


Aquatic invertebrate abundance across salinity gradients
Copepod
Ostracod
Cladocera
Hemiptera Corixidae Boatman
Annelida Oligochaete
Diptera Chironomid
Diptera Ceratopogonidae
Nematode
Annelida Polychaete
log + 1 transformed average abundance/L
Median salinity (ppt)